

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

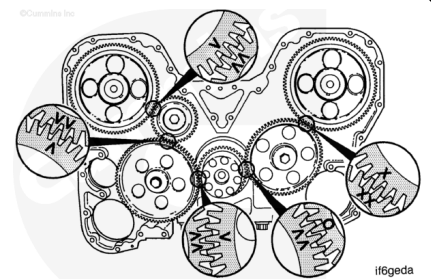
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Remove the front gear cover and all related components. Refer to Procedure 001-031 (</qs3/pubsys2/xml/en/procedures/28/28-001-031-tr.html>) in Section 1.

Remove

⚠ CAUTION ⚠

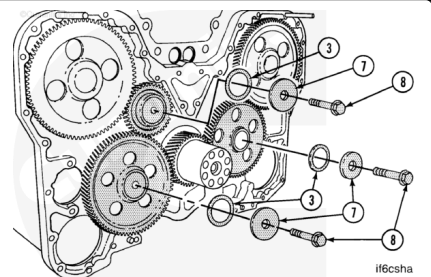
The three idler gears, two camshaft gears, and crankshaft gear have index marks that are aligned at assembly. Improper alignment will cause extensive engine damage.



Mark the gears to make sure they are reassembled in the proper position. Mark the gears to show if the engine has been barred over while the camshafts were removed.

⚠ CAUTION ⚠

Do not allow the gears or shafts to fall. Damage to the parts will result.



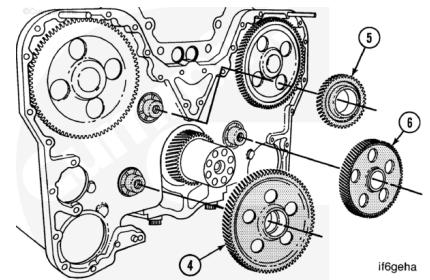
Note : It is recommended that the parts for each gear and shaft assembly be kept together to aid future assembly.

Note : It is recommended that the gear lash and the end clearance be measured before removal to identify any worn gears or thrust bearings.

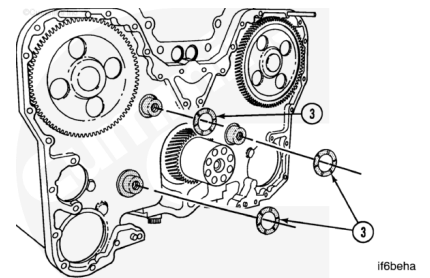
Remove the three capscrews (8). Remove the three retainers (7). Remove the three outer thrust bearings (3).

Remove the following:

- Water pump idler gear (4)
- Right bank camshaft idler gear (5)
- Left bank camshaft idler gear (6).



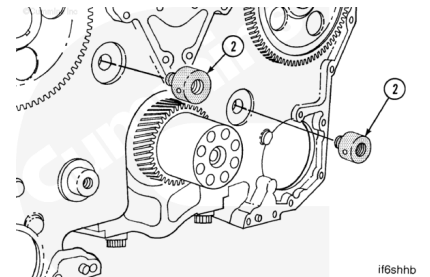
Remove the three inner thrust bearings (3).



Note : A slide hammer type tool that contains [3/4x16 inch) threads can be used to remove the idler shafts.

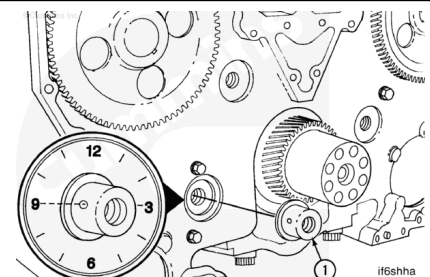
Remove the two camshaft idler gear shafts (2).

Carefully check the shafts and the bore in the cylinder block for any indication of movement from the shaft.



Check to see if the water pump idler shaft (1) has remained in alignment as shown (oil supply hole at 9 o'clock position). If there is any indication that the shaft has moved, be sure to check the parts thoroughly.

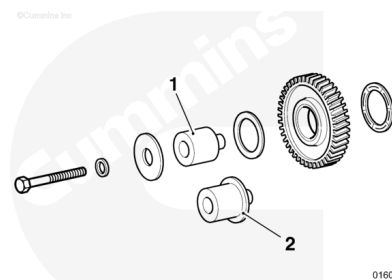
Remove the shaft.



Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

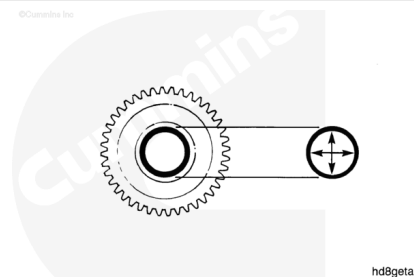


The shafts for camshaft idler gears and the water pump idler gears are different. The camshaft idler shafts (1) do **not** have an integral flange like the water pump idler shaft (2).

Clean and check the parts for damage.

Measure the inside diameter.

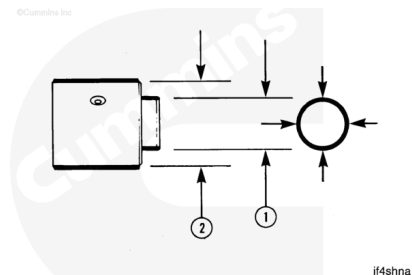
Idler Gear Bushing Inside Diameter		
mm		in
47.638	MIN	1.8755
47.714	MAX	1.8785



The bushing in the gear is precision bored after installation. If machining capability is **not** available, replace the bushing and the gear as an assembly.

⚠ CAUTION ⚠

Mounting parts for the water pump idler shaft are changed on newer engines. This flanged capscrew can be used if proper clearance and torque values are also applied.



Note : Some engines have a camshaft idler shaft that is the same as the water pump idler shaft.

Measure the outside diameter of the camshaft idler shaft.

Camshaft Idler Shaft Outside Diameter - (1)		
mm		in
25.387	MIN	0.9995
25.400	MAX	1.0000

Camshaft Idler Shaft Outside Diameter - (2)		
mm		in
47.549	MIN	1.8720
47.600	MAX	1.8740

Measure the outside diameter of the water pump idler shaft.

(1) - Engine Serial Number Higher than 33110701

mm		in
43.167	MIN	1.6995
43.180	MAX	1.7000

(2) - Engine Serial Number Higher than 33110701

mm		in
47.549	MIN	1.8720
47.600	MAX	1.8740

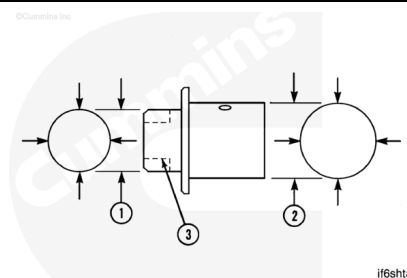
(3) - Engine Serial Number Lower than 33110701

mm		in
25.397	MIN	0.9995
25.400	MAX	1.0000

(2) - Engine Serial Number Lower than 33110701

mm		in
47.549	MIN	1.8720
47.600	MAX	1.8740

Note : Some engines with an engine serial number (ESN) lower than 33110701 have the shaft with the larger diameter (1). These engines were modified in the field until a service tool is available to modify the cylinder block.

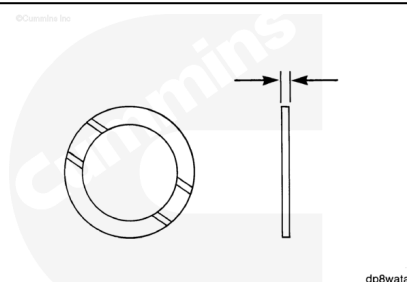


Check the grooved side for damage.

Measure the thickness.

Thrust Bearing thickness

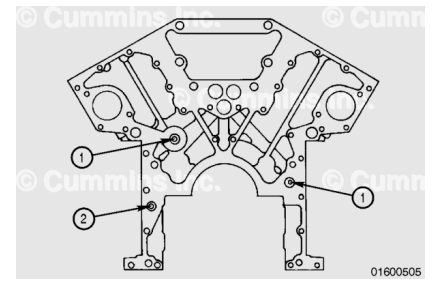
mm		in
2.34	MIN	0.092
2.39	MAX	0.094



Measure the inside diameter of the bore for the camshaft idler shafts (1) and water pump idler shaft (2).

Refer to Procedure 001-026

(/qs3/pubsys2/xml/en/procedures/28/28-001-026-tr.html) in
Section 1.



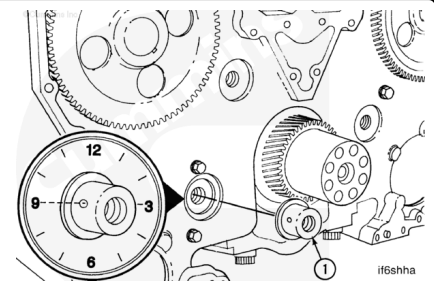
Install

⚠ CAUTION ⚠

Newer and older engines do not contain the same parts. If the correct parts are not installed, the engine will be seriously damaged.

⚠ CAUTION ⚠

The water pump idler shaft (1) must be installed so that the holes on the outside diameter of the shaft are at the 3 o'clock and 9 o'clock positions. If the holes of the shaft are not in the correct position, failure will result due to lack of lubrication of the bushing. It is recommended that the two camshaft idler shafts be installed with the holes in the same position as the water pump idler shaft.



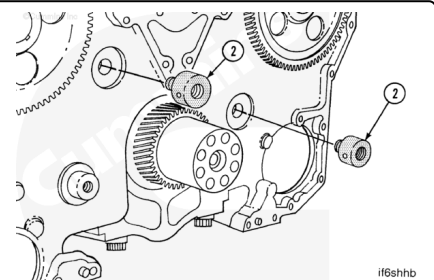
Mounting parts for the water pump idler shaft have changed on newer engines. This flanged capscrew can be used if proper clearance and torque values are applied.

Install the water pump idler shaft (1).

Rotate the shaft until the oil holes are at the 3 o'clock and 9 o'clock positions as shown.

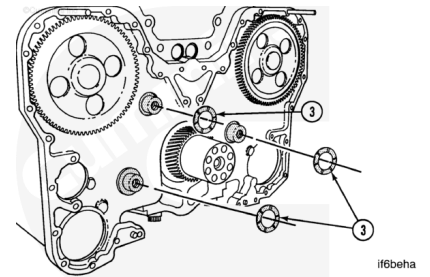
Install the two camshaft idler shafts (2).

Note : If a camshaft idler shaft does not stay in the block, check to be sure the parts are within specification.



⚠ CAUTION ⚠

The grooves on all thrust bearings (3) must be toward the idler gears. If the parts are installed wrong, failure will result.

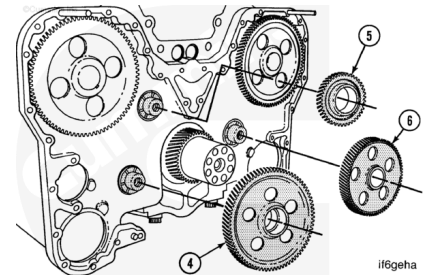


Use Lubriplate™ multi-purpose lubricant to lubricate the thrust bearings.

Install one thrust bearing (3) on each idler shaft.

⚠ CAUTION ⚠

The three idler gears, two camshaft gears, and crankshaft gear have index marks that must be aligned. If the marks are not aligned, the crankshaft and camshafts will not be timed properly and the engine will be extensively damaged.



Use Lubriplate™ multi-purpose lubricant to lubricate the bushings in the gears and the shafts.

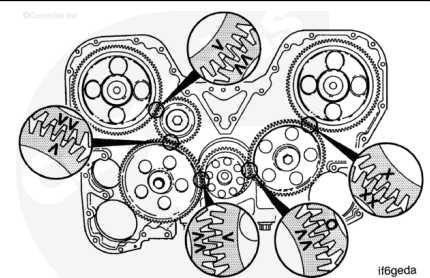
Install the following:

(4) Water pump idler gear

(5) Right bank camshaft idler gear

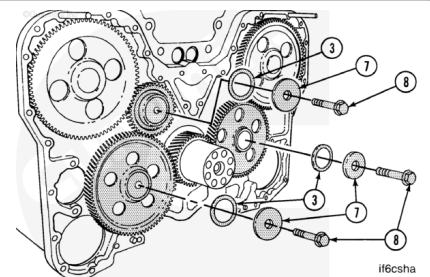
(6) Left bank camshaft idler gear.

Align the marks that were made during disassembly.



⚠ CAUTION ⚠

The grooves on the thrust bearings (3) must be toward the idler gears. If the bearings are installed wrong, failure will result

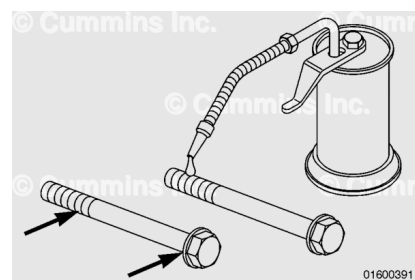


Use Lubriplate™ multi-purpose lubricant to lubricate the thrust bearings.

Install one thrust bearing (3) on each idler shaft.

Use clean engine oil. Lubricate the idler shaft capscrew threads and the flange head as shown.

Note : The torque method for the camshaft idler and the water pump idler capscrews are different.

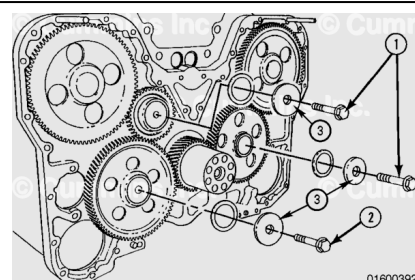


Install one thrust bearing retainer (3) and capscrew (1) on each camshaft idler shaft.

Torque Value:

Capscrews with flange head:

1. 205 N·m [151 ft-lb]
2. Loosen
3. 290 N·m [214 ft-lb]



Torque Value:

Capscrew without flange head: 250 N·m [184 ft-lb]

Install one thrust bearing retainer (3) and capscrew (2) onto the water pump idler shaft.

Note : There are two different bolt options for capscrew (2).

Option 2.0: Factory fitted prior to 15 August 2017. Bolt, Part Number 3177443. Bolt head size across flats 23.622 mm [0.930 in], 5/8 inch bolt.

Option 2.1: Factory fitted from 15 August 2017. Bolt, Part Number 3643905. Bolt head size across flats 27.94 mm [1.100 in], ¾ inch bolt.

Note option 2.1 may have been up-fitted to older blocks.

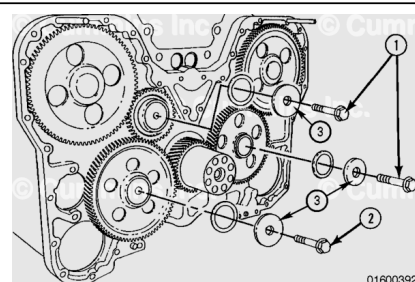
Always measure bolt head to identify the correct tightening method.

Torque Value:

Option 2.0 5/8 inch capscrews with flanged head:

1. 205 N·m [151 ft-lb]
2. Loosen
3. 88 N·m [65 ft-lb]
4. Rotate 90 degrees.

Note : In cases where the torque turn method is not possible due to limited access, a torque value of 305 N·m [225 ft-lb] can be used in place of the torque turn procedure. The torque only method must only be used in extreme cases.



Torque Value:

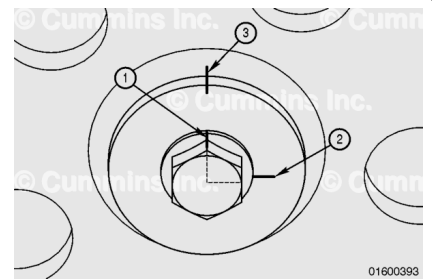
Capscrew without flange head: 250 N•m [184 ft-lb]

To apply the 90 degree angle for the water pump idler capscrew with a flanged head, perform the following:

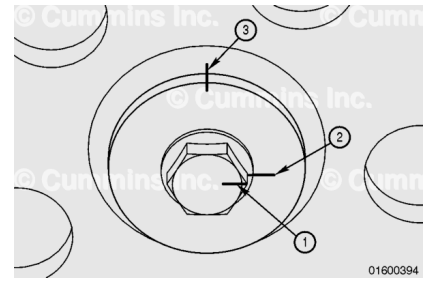
Once step 3 is complete (Torque Value: 88 N•m [65 ft-lb]), use a clean cloth to wipe excess oil from the thrust bearing retainer and capscrew head.

Use a marker to mark the capscrew (1), thrust bearing retainer (2), and thrust bearing retainer and idler gear (3) as indicated.

Note : Mark the capscrew at one of the points of the hexagon head (1). The 90 degrees will be located midway on the second flat in a **clockwise** direction from the marked point as shown in the diagram.

**⚠ CAUTION ⚠**

Make certain the thrust bearing retainer does not turn while rotating the capscrew through the 90 degrees. This will cause the bolt to over-torque and can result in engine failure. If the thrust bearing retainer turns, loosen the capscrews and start the torque procedure over.



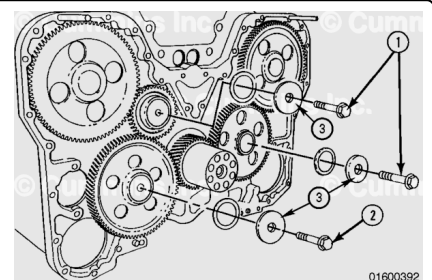
Turn the capscrew clockwise through 90 degrees until the mark on the capscrew (1) meets the mark on the thrust bearing retainer (2).

Note : If a dial indicator gauge or digital angle gauge is available, use one of these to improve accuracy of the final torque angle. The use of a gauge can eliminate the need for marking the components.

Torque Value:

Option 2.1 3/4 inch capscrews (1, 2) without flanged head:

1. 389 N•m [287 ft-lb]
2. Loosen
3. 271 N•m [200 ft-lb]
4. Rotate 90 degrees.

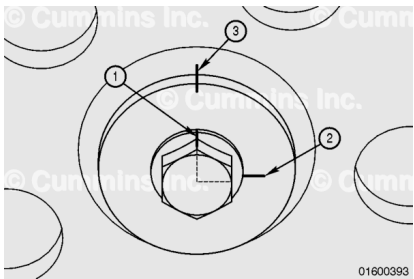


To apply the 90 degree angle for the water pump idler capscrew without a flanged head, perform the following:

Once step 3 is complete (Torque Value: 271 N•m [200 ft-lb]), use a clean cloth to wipe excess oil from the thrust bearing retainer and capscrew head.

Use a marker to mark the capscrew (1) thrust bearing retainer (2), and thrust bearing retainer and idler gear (3) as indicated.

Note : Mark the capscrew at one of the points of the hexagon head (1). The 90 degrees will be located midway on the second flat in a clockwise direction from the marked point as shown in the diagram.

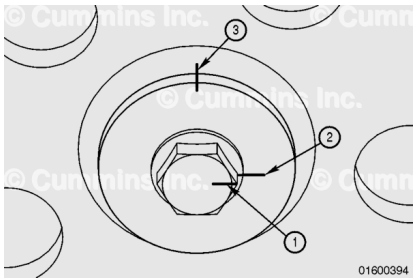


⚠ CAUTION ⚠

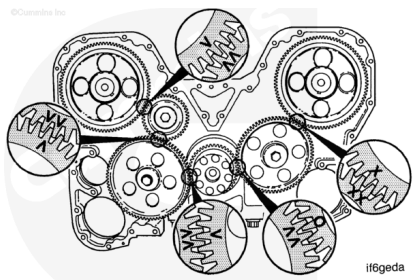
Make certain the thrust bearing retainer does not turn while rotating the capscrew through the 90 degrees. This will cause the bolt to over-torque and can result in engine failure. If the thrust bearing retainer turns, loosen the capscrews and start the torque procedure over.

Turn the capscrew clockwise through 90 degrees until the mark on the capscrew (1) meets the mark on the thrust bearing retainer (2).

Note : If a dial indicator gauge or digital angle gauge is available, use one of these to improve accuracy of the final torque angle. The use of a gauge can eliminate the need for marking the components.



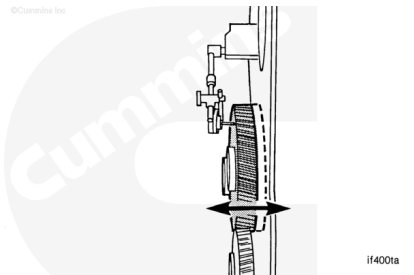
Check the alignment marks again to be sure that they are aligned as shown.



Use a dial indicator to measure the end clearance.

Note : The end clearance measurements are the same for the newer and older idler assemblies.

Idler Gear End Clearance		
mm		in
0.13	MIN	0.005
0.46	MAX	0.018



If the clearance is **not** within specifications, check for foreign material between the parts.

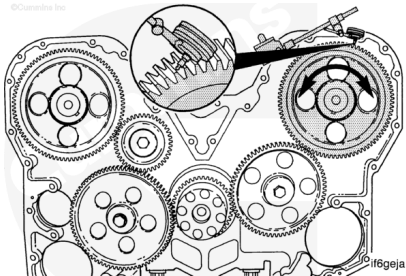
If the clearance is **not** within specifications, check to see if the thrust bearings are in the proper locations. Oversize thrust bearings are available.

Measure the Front Gear Train Backlash

Note : The adjacent (engaging) gear must not turn as the gear that is being measured is turned.

Use a dial indicator. Position the indicator on a gear as shown. Measure the backlash on the gears.

Rotate the gear being measured **clockwise**. Position the indicator at “0”. Rotate the gear **counterclockwise**. Read the indicator.



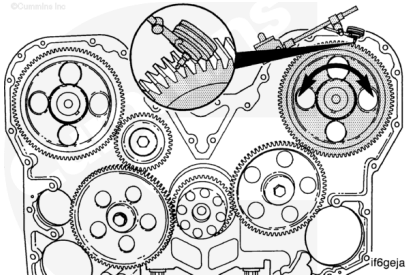
Front Gear Train Gear Backlash

mm		in
0.07	MIN	0.003
0.51	MAX	0.020

Repeat this procedure for all the gears except the crankshaft gear.

If the backlash is **not** within specifications, be sure the adjacent (engaging) gear is **not** moving during the measurement.

If the backlash is **not** within specifications, the gear **must** be replaced.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the gear cover and all related components. Refer to Procedure 001-031 (/qs3/pubsys2/xml/en/procedures/28/28-001-031-tr.html) in Section 1.

- Operate the engine. Check for leaks.

Last Modified: 11-Dec-2019
